Day 02: Gaussian Elimination, Fundamental Matrices, and Column Space

1. The Gaussian Elimination Process ($Ax = b$)

Gaussian Elimination is the algebraic "engine" used to solve the linear system $Ax = b$. Instead of just looking at the matrix A, we apply operations to both A and the target vector b to transform the system into an **Upper Triangular form** ($Ux = c$).

1.1 Step-by-Step Numerical Example

Let's solve the system $Ax = b$ where:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Step 1: Eliminate elements below the first pivot ($A_{11} = 2$).

- To eliminate the 4 in Row 2, we subtract 2 times Row 1 from Row 2 ($R_2 \leftarrow R_2 - 2R_1$).
  - Row 2 of A becomes: $[4 - 2(2), -6 - 2(1), 0 - 2(1)] = [0, -8, -2]$.
  - b_2 becomes: $-2 - 2(5) = -12$.
- To eliminate the -2 in Row 3, we add Row 1 to Row 3 ($R_3 \leftarrow R_3 + R_1$).
  - Row 3 of A becomes: $[-2 + 2, 7 + 1, 2 + 1] = [0, 8, 3]$.
  - b_3 becomes: $9 + 5 = 14$.

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -12 \\ 14 \end{bmatrix}$$

Step 2: Eliminate elements below the second pivot ($A_{22} = -8$).

- To eliminate the 8 in Row 3, we add Row 2 to Row 3 ($R_3 \leftarrow R_3 + R_2$).
  - Row 3 of A becomes: $[0 + 0, 8 + (-8), 3 + (-2)] = [0, 0, 1]$.
  - b_3 becomes: $14 + (-12) = 2$.

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -12 \\ 2 \end{bmatrix}$$

Step 3: Back-substitution to find the solution.

Now that we have the Upper Triangular Matrix U, we solve from the bottom up: 1. From R_3: $1z = 2 \implies \mathbf{z = 2}$. 2. From R_2: $-8y - 2(z) = -12 \implies$

$-8y - 4 = -12 \implies -8y = -8 \implies \mathbf{y = 1}$. 3. From R_1: $2x + 1(y) + 1(z) = 5 \implies 2x + 1 + 2 = 5 \implies 2x = 2 \implies \mathbf{x = 1}$.

The final state vector is $\mathbf{x} = [1, 1, 2]^T$.

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2. Row Operations as Matrix Multiplication

Every step in elimination is actually a matrix multiplication. If we subtract 2 times Row 1 from Row 2, we are multiplying A by an **Elementary Matrix** (E_{21}).

To create E_{21}, start with the Identity matrix and put the multiplier (with a negative sign) in the target position:

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{-2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Applying all steps as matrices:

$$E_{32}(E_{31}E_{21})A = U$$

In robotics, this is critical because we often represent motion as a sequence of matrix multiplications. Solving equations is just a special case of this transformation.

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3. Fundamental Matrices in Robotics

3.1 Identity Matrix (I)

The I matrix is the "1" of linear algebra. It represents "no change."

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \implies IA = A$$

3.2 Permutation Matrix (P)

If a pivot is 0, we must swap rows. P is an Identity matrix with reordered rows. To swap Row 1 and Row 2:

$$P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \implies P_{12} \begin{bmatrix} \text{Row 1} \\ \text{Row 2} \\ \text{Row 3} \end{bmatrix} = \begin{bmatrix} \text{Row 2} \\ \text{Row 1} \\ \text{Row 3} \end{bmatrix}$$

3.3 Matrix Transpose (A^T)

The transpose flips the matrix over its diagonal (A_{ij} becomes A_{ji}).

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \implies A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Robotics Insight: For rotation matrices R, the transpose R^T is exactly the inverse R^{-1} (the reverse rotation).

3.4 Matrix Inverse (A^{-1})

The inverse "undoes" the matrix. If $Ax = b$, then $x = A^{-1}b$.

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \implies A^{-1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{(Since } AA^{-1} = I\text{)}$$

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4. Success and Failure: The Geometry of Column Space

A robot's configuration is valid only if the target motion b is reachable by its actuators (the columns of A).

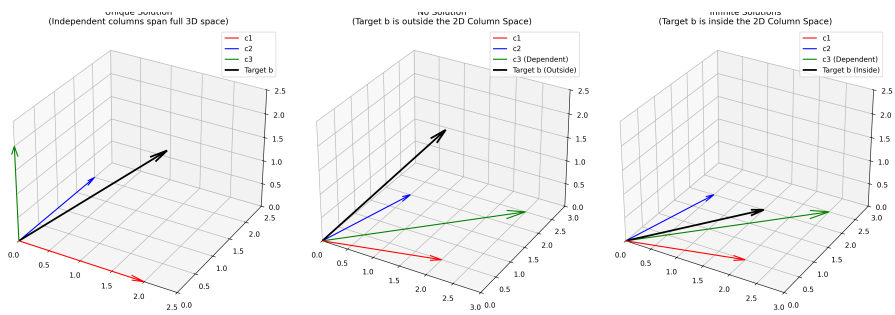

Figure 1: Column Space geometry: spanning 3D space vs 2D plane

1. **Unique Solution:** The columns c_1, c_2, c_3 are independent. They span all of 3D space. **Any** target b can be reached.
2. **No Solution:** The columns are dependent (e.g., $c_3 = c_1 + c_2$). They span only a **2D plane**. If the target b is outside this plane, the robot cannot reach it.
3. **Infinite Solutions:** The columns span a 2D plane, and b happens to be inside that plane. There are many ways to reach the same point.

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5. Quick Quiz

1. **Inference:** If a 3x3 matrix has only two non-zero pivots after elimination, what is the dimension of its Column Space?
2. **Matrix Logic:** If E_{21} subtracts 2 times Row 1 from Row 2, what does the inverse matrix E_{21}^{-1} look like? (Hint: It should add 2 times Row 1 back).
3. **Robotics:** Why do we say a robot is "Singular" when its Jacobian matrix loses a pivot during elimination?

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6. References

- **MIT 18.06 Linear Algebra** (Prof. Gilbert Strang), Lecture 03.
- **UMich ROB 501: Mathematics for Robotics**, Matrix Algebra & Systems of Equations.
- **Open Tutorial Project**: Linear Algebra for Robotics.

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7. Python Implementation

```python
import numpy as np

# Define A and b from the numerical example
A = np.array([[2, 1, 1],
              [4, -6, 0],
              [-2, 7, 2]], dtype=float)
b = np.array([5, -2, 9])

# 1. Solving Ax = b directly
x = np.linalg.solve(A, b)
print(f"Computed Solution x: {x}")

# 2. Creating a Permutation Matrix to swap Row 1 and Row 2
P = np.array([[0, 1, 0],
              [1, 0, 0],
              [0, 0, 1]])
print("\nSwapped Matrix A (P*A):")
print(P @ A)

# 3. Computing the Transpose
print("\nTranspose of A:")
print(A.T)

# 4. Checking if Matrix is Singular (failure case)
```

```python
A_singular = np.array([[1, 1, 1], [2, 2, 2], [0, 1, 0]])
rank = np.linalg.matrix_rank(A_singular)
if rank < 3:
    print("\nWarning: Matrix is singular. It only spans a 2D space!")
```